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Stabilizing 1.93-eV Ultrawide-Bandgap Perovskites for Efficient Triple-Junction Solar Cells

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Abstract

Monolithic perovskite/perovskite/silicon triple-junction solar cells offer higher theoretical efficiency than their two-junction counterparts. However, the phase instability of ultrawide-bandgap perovskites remains a major obstacle to further improve the efficiency and stability of triple-junction devices. Here, we introduce 1,4-phenylenediamine dihydriodide into the 1.93-eV bandgap perovskite, which significantly increases the ion migration barrier and suppresses phase segregation through strong interactions with the perovskite lattice. Combined with a post-treatment using 3-(methylthio)propylamine hydroiodide to modify the perovskite surface, we achieved a certified efficiency of 26.18% for the triple-junction solar cell, with an impressive open-circuit voltage of 3.148 V on an aperture area of 1.17 cm². Notably, the encapsulated devices retained 96% of their initial performance after 1500 hours of continuous maximum power point tracking under one-sun illumination in ambient conditions, representing the notable stability in perovskite/perovskite/silicon triple-junction cells reported to date.

Introduction

Multi-junction solar cells have great potential to further enhance power conversion efficiency (PCE) by stacking multiple light-absorbing layers with complementary bandgaps, compared with single-junction devices. The theoretical Shockley-Queisser limit for perovskite/perovskite/silicon triple-junction (3-J) solar cells is 49.4%, exceeding the 45.1% limit for perovskite/silicon two-junction (2-J) cells^{1,2}. However, the reported highest certified PCE of perovskite/perovskite/silicon cells is only 27.1%, far below the 34.6% PCE of perovskite/silicon cells at present^{3,4}. In fact, when the bromide fraction exceeds 20% in perovskite compositions, the resulting wide-bandgap perovskites exhibit non-radiative recombination losses due to high defect densities and increased halide segregation under illumination. Specially, for ultrawide-bandgap perovskites with bromide fractions above 50% in 3-J solar cells, these issues become more pronounced, as the higher Br/I ratio exacerbates defect formation and halide phase separation, leading to reduced PCE and instability during various device operation conditions. Therefore, the primary challenge to realizing stable and efficient 3-J solar cells arises from the intrinsic phase heterogeneity of ultrawide-bandgap perovskites (1.9-2.0 eV)³⁻¹¹. Secondly, the high-density defects at the interface are also an important factor in PCE loss and performance degradation of perovskite solar cells, and for intrinsically high-defect-density ultrawide-bandgap perovskites, the efficiency and stability issues caused by interface defects are even more severe.

Shi *et al.* reported that gas quenching method effectively modulates the crystallization of wide-bandgap perovskite films by elevating the nucleation energy barrier, leading to smooth films characterized

by a uniform vertical halogen distribution¹². Wang *et al.* introduced the small-radius A-site cation Rb⁺ into Rb⁺/Cs⁺ mixed-cation inorganic 2.0 eV perovskites, which increased the ion-migration energy barrier by reducing the interatomic distance between the A-site cation and iodide ions, thereby suppressing phase segregation¹³. In addition, replacing halide ions at the X-site with pseudo-halide ions has also been explored. The cyanate not only improves the uniformity of I/Br distribution and increases the formation energy of vacancy defects, but also regulates crystal growth^{3,14}. In addition, ammonium halides not only stabilize wide-bandgap perovskite solar cells, but are also widely used to reduce defect density and energy level matching¹⁵⁻¹⁸. Therefore, to address the stability and efficiency challenges of 1.9-2.0 eV perovskites, incorporating the ammonium salts into the perovskite composition to suppress halide phase separation and passivating defects in both the bulk and at the interfaces of the perovskite layer may be an effective strategy for improving phase stability and enhancing device performance. Khadka *et al.* demonstrated that the diammonium iodide molecules—including PEDAI (C₆H₈N₂*2HI) and PZDI (C₄H₁₀N₂*2HI)—affect the film morphology and surface chemistries in FA_{0.84}Cs_{0.12}Rb_{0.04}PbI₃ perovskite via their basicity and adsorption energy¹⁹. Separately, the performance of the device has been improved by depositing a mixed solution of 3-(methylthio)propylamine hydrochloride (3MTPAI) and propane-1,3-diammonium iodide (PDAI₂) on Cs_{0.05}MA_{0.05}FA_{0.90}PbI₃ perovskite film²⁰. However, the effects and mechanisms of PEDAI and 3MTPAI on the performance of ultrawide bandgap perovskite remain unexplored, separately.

In this work, we incorporate 1,4-phenylenediamine dihydriodide (PEDAI) into a 1.93-eV ultrawide-bandgap perovskite composition. The strong interaction between PEDAI⁺ and [PbX₆]⁴⁻ stabilize the perovskite lattice, increasing the energy barrier for ion migration, thereby significantly enhancing phase stability under various aging conditions. In combination with an interfacial treatment using 3MTPAI to passivate interface defects and optimize interface energy alignment, we achieved a single-junction efficiency of 16.09% for 1.93-eV bandgap perovskite solar cells. Furthermore, a certified efficiency of 26.18% and an impressive *V*_{oc} of 3.148 V were obtained for the perovskite/perovskite/silicon triple-junction solar cells. The encapsulated 3-J devices retained over 96% of their initial PCE after 1500 hours of continuous maximum power point tracking under simulated 1-sun illumination in ambient conditions, representing the best stability in 3-J solar cells reported to date.

Results

Photostability of 1.93-eV perovskite films

The perovskite composition Cs_{0.25}FA_{0.6}MA_{0.15}Pb(I_{0.43}Br_{0.57})₃ is employed as an optimal top absorber layer in the 3-J solar cells, owing to its favorable 1.93-eV bandgap. However, such high Br/I ratios are

prone to spontaneous phase segregation¹⁹, as evidenced by the photoluminescence (PL) spectra of the control film shown in Fig. 1a. Under light illumination, the dominant peak of the $\text{Cs}_{0.25}\text{FA}_{0.6}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.43}\text{Br}_{0.57})_3$ species gradually redshifts from 652 nm to 688 nm, accompanied by the emergence of an additional peak at 725 nm, which further shifts to 761 nm. These spectral evolutions reveal the formation of photoinduced, I-rich phase-segregated domains driven by halide ion migration, which would ultimately lead to poor phase stability and a deficit open-circuit voltage (V_{oc}) in solar cells²¹⁻²³. Moreover, the overall PL intensity of the control film increases substantially after light illumination, suggesting that the newly formed I-rich components exhibit higher luminescence efficiency than the original $\text{Cs}_{0.25}\text{FA}_{0.6}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.43}\text{Br}_{0.57})_3$ perovskite films. This enhancement can be attributed to the I-rich phase generated during light soaking, which possesses high defect tolerance and thus facilitates radiative recombination of electrons and holes²⁴⁻²⁶. However, this phenomenon also leads to unfavorable bandgap matching with the other two perovskite sub-cells that have different bandgaps, which hinders the fabrication of high-efficiency 3-J solar cells. To efficiently suppress halide phase segregation in this perovskite composition, PEDAI (structure shown in Supplementary Figure 1) was incorporated into the perovskite layer. PL measurements show that no spectral shifts or additional emissive species were observed after continuous illumination for over 10 minutes (Fig. 1b, c). Similarly, as the excitation intensity increased, phase separation was observed in the control film, while the PEDAI film remained stable under the same conditions (Supplementary Figure 2). Scanning electron microscope (SEM) characterization (Fig. 1d, e) further highlights the photostability of the PEDAI-treated perovskite films, which preserve their morphology after 48 h of continuous 1-sun illumination, whereas the control films display pinholes and compositional heterogeneity. Consistent with these observations, X-ray diffraction (XRD) analysis (Fig. 1f and Supplementary Figure 3a) reveals pronounced structural degradation in the control film upon light exposure. Specifically, the intensity of the (100) diffraction peak at approximately 14.7° decreased markedly to 57% of its initial value after 24 h, accompanied by peak broadening from 0.17° to 0.30° , indicating of increased structural disorder and phase-segregation-induced degradation of the perovskite lattice²⁷⁻²⁹. After 48 h of continuous illumination, the peak intensity further declined to 50%, with additional broadening to 0.39° . In contrast, the PEDAI-treated perovskite films exhibited only minor reductions in (100) peak intensity and negligible broadening under the same aging conditions (Fig. 1f and Supplementary Figure 3b). These results indicate that the introduction of PEDAI effectively suppresses ion migration and mitigates phase segregation, thereby enhancing the phase stability of the 1.93-eV perovskite under light illumination.

Stabilization mechanism

Beyond light-induced phase stability, the phase stability of PEDAI-treated perovskite films was further assessed under accelerated thermal aging in ambient air. When the as-prepared control film was heated at elevated temperatures (110-130 °C) under 30-40% relative humidity in air, a pronounced discoloration appeared within 20 min (Supplementary Figure 4a). Additionally, the film color progressively changed from brown to orange upon prolonged heating under different temperatures, indicating a phase change associated with an increased bromide-rich composition. In contrast, the PEDAI-treated perovskite film preserved its photoactive phase, exhibiting no noticeable color change for 60 minutes under the same conditions (Supplementary Figure 4b). These results can be further confirmed by temperature-dependent XRD measurements of the above films, as shown in Supplementary Figures 5 and 6. As the heated time increases, the XRD peak shifts to large angle for the control film (Supplementary Figure 5a-c), explaining the bromine-rich phase, whereas the PEDAI is not obvious (Supplementary Figure 6a-c). Accordingly, the extracted activation energies (E_a) of the control and PEDAI-treated films from above XRD (Fig. 1g, Supplementary Figure 7 and Supplementary Note 1), were determined to be 0.62 eV and 0.80 eV, respectively^{21,30,31}. This substantial increase in E_a indicates that PEDAI incorporation elevates the energy barrier for ion migration within the ultrawide-bandgap perovskite, thereby effectively suppressing phase decomposition in air and markedly enhancing phase stability³². The PEDAI film exhibited negligible shifts in its UV-vis absorption spectrum after 24 and 48 h of either continuous 1-sun illumination (Supplementary Figure 8) or thermal aging at 85 °C (Supplementary Figure 9), whereas the control film showed significant changes under the same conditions, further confirming the enhanced phase stability achieved with PEDAI.

To investigate the interaction between PEDAI and the perovskite composition, high-resolution X-ray photoelectron spectroscopy (XPS) was performed. As shown in the results, the characteristic Pb 4f_{7/2} and Pb 4f_{5/2} peaks of the control film, originally located at 138.1 and 143.0 eV, respectively, shifted to lower binding energies of 137.8 and 142.7 eV after PEDAI treatment (Fig. 1h). In addition, a corresponding shift in the I 3d and Br 3d peak was observed (Supplementary Figure 10), indicating a modified electronic environment, primarily resulting from strong hydrogen-bonding interactions between PEDAI and [PbX₆]⁴⁻³³⁻³⁵. The intermolecular interaction could efficiently increase the formation energy of halide vacancies, inhibiting ion migration during light and thermal conditions. Furthermore, time-of-flight secondary ion mass spectrometry (ToF-SIMS) elemental depth profiling confirmed the distribution of PEDAI⁺ species throughout the perovskite layer (Supplementary Figure 11), corroborating the effective incorporation of PEDAI at the molecular level. Notably, the progressive increase in PEDAI⁺ signal strength from the surface to the bottom is indicative of accumulation at the buried interface with the hole transport

layer. PEDAI contains two ammonium functional groups that can react with lead halides to form a 2D perovskite phase, as reported in previous studies¹⁹. However, no distinct low-angle diffraction peaks ($2\theta < 10^\circ$) were observed in the XRD or grazing incidence wide-angle X-ray scattering (GIWAXS) patterns of the PEDAI-treated films in this work (Fig. 1i and Supplementary Figure 3). To examine the reactivity of PEDAI with lead halide species, PEDAI was mixed with lead iodide with a 1:1 molar ratio. After annealing process, two low-angle peaks below 10° were clearly observed, confirming the formation of low-dimensional species¹⁹, as shown in Supplementary Figure 12. In contrast, no low-angle peaks appeared when mixing with lead bromide (PbBr_2), which might due to the poor crystallization properties of bromine-rich composition in Supplementary Figure 13; nevertheless, the disappearance of the PbBr_2 diffraction peaks suggests strong chemical interaction and high reactivity between PEDAI and PbBr_2 . To further investigate the interactions at the atomic and nanometer scales after PEDAI treatment, transmission electron microscopy (TEM) measurements were performed. Compared with the control films, no obvious low-dimensional crystalline structures were clearly observed in the PEDAI-treated films. Meanwhile, The PEDAI, containing a benzene ring, is too large to enter the crystal lattice within three-dimensional perovskite. However, compared with the control films, the interplanar spacing increases from 5.99 Å to 6.12 Å (Fig. 2a, b), accompanied by a shift of the XRD diffraction peaks toward lower 2θ angles in the PEDAI-treated films (Supplementary Figure 14), which also indicates that the PEDA^+ occupies the A-site. Despite the strong chemical interaction and high reactivity between PEDAI and PbBr_2 , and the incorporation of PEDA^+ into A-site, the lack of clearly detectable low-dimensional phases by XRD or GIWAXS can be primarily attributed to the low concentration of PEDAI and the poor crystallization property of bromide-rich the perovskite composition. Overall, the strong interaction between PEDAI and perovskite lattice efficiently suppress ion migration and inhibit the undesired phase segregation.

Optoelectronic properties of perovskite films

Subsequently, we investigated the influence of PEDAI incorporation on the photoelectronic properties of the perovskite films. The control film exhibits pronounced spatial PL heterogeneity and a lower overall PL intensity (Fig. 2c and e, Supplementary Figure 15a)^{36,37}. In contrast, the PEDAI-treated film (Fig. 2d and e, Supplementary Figure 15b) displays a strong and homogeneously distributed PL signal across the entire surface, suggesting a lower rate of trap-induced non-radiative recombination and a phase homogeneity. Additionally, the prolonged PL lifetime (Fig. 2f) and the substantial increase in quasi-Fermi level splitting (QFLS, Fig. 2g, derived from PL quantum yield (PLQY) measurements in Supplementary Table 1, see Supplementary Note 2 for details) observed in the PEDAI-treated perovskite film provide

compelling evidence that PEDAI effectively passivates defects and suppresses non-radiative recombination pathways³⁸⁻⁴¹.

Ultraviolet photoelectron spectroscopy (UPS) was conducted to investigate the influence of PEDAI treatment on the energy-level alignment of the perovskite layer. The valence band maximum value of PEDAI bottom film was closer to hole transport layer compared to control bottom film, showing an improved band alignment with hole transport (Supplementary Figure 16). As shown in Fig. 2h (Supplementary Figures 17 and 18), the incorporation of PEDAI induces negligible changes in the conduction band positions compared with the control surface perovskite film, which exhibit considerable perovskite/C₆₀ interfacial recombination loss⁴²⁻⁴⁵. Therefore, to further minimize this energy loss and enhance device performance, 3MTPAI (structure shown in Supplementary Figure 19) was introduced as a post-treatment on the PEDAI-treated perovskite surface, hereafter denoted as the bulk and interface engineered (BIE) sample. As shown in Fig. 2h (Supplementary Figures 17 and 18), the conduction bands minimum offset between the perovskite and C₆₀ decreased from 0.28 eV in the control film to 0.09 eV after 3MTPAI incorporation, indicating improved interfacial band alignment, thereby reducing the electron transfer barrier and promoting more efficient electron extraction. Meanwhile, the further increase in PL intensity, the prolonged carrier lifetime and excellent QFSL indicate that 3MTPAI treatment effectively suppresses interfacial defects between the perovskite and C₆₀ layers, as shown in Fig. 2e-g.

Photovoltaic performance and photostability of solar cells

Therefore, single-junction opaque solar cells incorporating PEDAI- and BIE-treated perovskite layers were fabricated in a p-i-n configuration with the architecture of glass/ITO/NiO_x/SAM/1.93-eV perovskite/C₆₀/SnO₂/Ag. As shown in Supplementary Figures 20-22, the PEDAI-treated film exhibited fewer grain boundaries, higher crystallinity, and much lower defect densities. These microstructural improvements greatly reduce shunt issues and voltage loss, leading to an increase in both *V*_{oc} and FF. Consequently, the overall performance of PEDAI devices is enhanced, surpassing that of devices with other diammonium salt additives (Supplementary Figures 23 and 24). Statistical box plots summarizing the photovoltaic parameters at optimized concentrations are shown in Supplementary Figures 25 and 26, with a direct comparison and presented in Fig. 3a. The hysteresis index (HI) values were derived by combining the forward and reverse scan PCE of the devices (Fig. 3a), with the statistical box plot shown in Supplementary Figure 27. The BIE devices demonstrate the most concentrated distribution and the lowest average HI values, indicating improved stability and excellent reproducibility. Meanwhile, the BIE-treated devices exhibited the highest PCE, driven primarily by notable enhancements in both *V*_{oc}

and FF, as shown by the J - V curves in Fig. 3b. The V_{oc} of the BIE devices increased by approximately 50 mV compared with the control, consistent with the QFSL results in Fig. 2g. Correspondingly, the average FF improved from ~74.01% to 81.33%. Consequently, the champion BIE-treated device achieved an impressive PCE of 16.09%, with a V_{oc} of 1.395 V, J_{sc} of 14.08 mA cm⁻², and FF of 81.91% (Supplementary Table 2), whereas the best control device reached a PCE of 14.14% with a V_{oc} of 1.348 V, J_{sc} of 13.81 mA cm⁻², and FF of 75.93%. Furthermore, integration of the external quantum efficiency (EQE) spectra weighted by the AM 1.5G solar spectrum (Fig. 3c) yielded current densities of 13.89 and 14.11 mA cm⁻² for the control and BIE devices, respectively, in good agreement with the J - V measurements. The enhanced photocurrent can be attributed to the improved crystal quality of the BIE-treated perovskite film. Meanwhile, it is observed that under maximum power point tracking (MPPT) for 300 s under one-sun illumination, the PEDAI- and BIE-treated device maintained stable output, whereas the control device exhibited noticeable degradation (Fig. 3d). The BIE device retained 96% of its initial efficiency after a UV dose of 10 kWh/m², while the control device exhibited faster degradation (Supplementary Figure 28), highlighting the improved UV stability achieved with BIE treatment.

Additionally, the carrier recombination behavior of the devices under different treatments was investigated by analyzing the dependence of V_{oc} on light intensity (Fig. 3e). The semilogarithmic relationship follows the expression with a slope ($nkT/q \log_{10}e$), where n is the diode quality factor^{37,46}. The extracted n values for the control, PEDAI-, and BIE-treated devices were 1.750, 1.611, and 1.507, respectively, indicating that the BIE-treated devices exhibit the lowest trap-assisted recombination losses. To further probe the energetic disorder, the Urbach energy (E_U), which reflects the degree of disorder and defect states in perovskite films, was determined using Fourier-transform photocurrent spectroscopy (FTPS)^{46,47}. As shown in Supplementary Figure 29, the optical bandgap remained nearly unchanged after surface modification. However, the average E_U decreased from 28.59 meV for the control devices to 24.09 meV and 21.45 meV for the PEDAI- and BIE-treated devices, respectively (Fig. 3f and Supplementary Figure 30). The reduced E_U for the BIE-treated samples indicates diminished defect densities and suppressed energetic disorder, consistent with the light-intensity-dependent V_{oc} results. Moreover, electrochemical impedance spectroscopy (EIS) further corroborated these findings. The Nyquist plots (Fig. 3g), obtained under dark conditions with an applied bias of 1.05 V over a frequency range of 0.1 MHz to 1 Hz⁴⁸, revealed that the BIE-treated devices exhibited a lower series resistance (R_s) and a higher recombination resistance (R_{rec}) compared to the control devices (Supplementary Table 3). This trend further confirms that BIE treatment can effectively reduce charge recombination and enhance carrier transport in solar cell⁴⁹.

Encouraged by the excellent performance and photostability of the BIE-treated devices, we further developed perovskite/perovskite/silicon triple-junction solar cells. To maximize the overall PCE, a 1.52-eV perovskite absorber with the composition $\text{Cs}_{0.05}\text{FA}_{0.85}\text{MA}_{0.1}\text{PbI}_3$ was employed for the middle sub-cell, ensuring optimal bandgap alignment with the 1.93-eV top sub-cell. Single-junction opaque devices based on the 1.52-eV perovskite, configured as glass/ITO/ NiO_x /SAM/1.52-eV perovskite/ C_{60} /SnO₂/Ag, achieved a champion PCE of 23.67% with a J_{sc} of 25.24 mA cm⁻² (Supplementary Figure 31). Subsequently, 1.52-eV perovskite/silicon half-cells were fabricated to assess integration feasibility. Top-view and cross-sectional SEM images, along with J - V and EQE characteristics of the textured heterojunction silicon (HJT) bottom cell, are presented in Supplementary Figures 32 and 33. The SEM images of the resulting 2-J half-cells (Supplementary Figure 34) reveal uniform perovskite coverage on the textured silicon surface with feature sizes below 1 μm . These 2-J devices delivered a PCE of 24.22% with a J_{sc} of 16.59 mA cm⁻² (Supplementary Figure 35), providing an ideal platform for subsequent deposition of the ultrawide-bandgap perovskite top absorber.

Finally, we integrated the 1.93-eV perovskite top absorber with the previously developed 1.52-eV perovskite/silicon half-cell to construct a monolithic Si/ITO/ NiO_x /SAM/1.52-eV perovskite/ C_{60} /SnO₂/ITO/ NiO_x /SAM/1.93-eV perovskite/ C_{60} /SnO₂/IZO/Ag 3-J solar cell. A schematic illustration of device architecture is presented in Fig. 4a. The cross-sectional SEM images (Fig. 4b) clearly identify the individual sub-cells: the 1.52-eV perovskite layer fully conforms to the textured silicon surface, while the 1.93-eV layer forms a smooth, uniform top film. Photograph of the 2.5 cm * 2.5 cm perovskite/perovskite/silicon device is shown in Fig. 4c. Statistical performance data for the control and BIE-treated devices are summarized in Supplementary Figure 36. The champion 3-J device achieved a PCE of 26.62%, with a V_{oc} of 3.130 V, J_{sc} of 10.96 mA cm⁻², and FF of 77.63% (Supplementary Figure 37). Independent certification performed by the Shanghai Institute of Microsystem and Information Technology (SIMIT) confirmed a reverse-scan PCE of 26.18% (Fig. 4d and Supplementary Figure 38), accompanied by a high open voltage of 3.148 V in 3-J cells (Supplementary Figure 39)^{3,15,16,50-54}. EQE-derived current densities for the top, middle, and bottom sub-cells were 12.26, 11.07, and 14.50 mA cm⁻², respectively (Fig. 4e), indicating that the middle perovskite sub-cell limits the overall J_{sc} of the triple-junction device. Therefore, narrowing the bandgap of the middle perovskite sub-cell can further boost the efficiency of triple-junction tandem cells. Under continuous MPPT for 800 s, the device retained a stabilized efficiency of 25.85% (Fig. 4f). Furthermore, to evaluate operational durability, we monitored the MPPT stability of encapsulated triple-junction devices under AM 1.5 G illumination in ambient air (Fig. 4g and Supplementary Figure 40). The control devices degraded rapidly, dropping to 82% of initial

PCE within just 110 hours in Supplementary Figure 40a. In contrast, the PCE of encapsulated BIE triple-junction solar cells maintained 96% of the initial PCE (25.2%) over 1500 hours of continuous operation. This remarkable stability is primarily attributed to the incorporation of PEDAI, as devices incorporating PEDAI alone retained their initial PCE of 24.4% without any noticeable degradation over 1,500 hours (Supplementary Figure 40b). To the best of our knowledge, these results represent the best reported value to date for perovskite/perovskite/silicon and perovskite/perovskite/perovskite triple-junction solar cells (Fig. 4h)^{3,15,18,53-56}. This impressive combination of high efficiency and long-term stability underscores the strong potential of silicon-based triple-junction perovskite photovoltaics.

Discussion

In this work, we introduced PEDAI into a ultrawide-bandgap perovskite composition, forming strong interactions with the perovskite lattice that effectively suppress undesired phase segregation in 1.93-eV perovskites composition. After the combination of 3MTPAI interface treatment reduced the defect density and matched the energy levels between perovskite and C₆₀, we successfully fabricated perovskite/perovskite/silicon triple-junction cells achieving over 26.18% (1.17 cm²) PCE with open-circuit voltage of 3.148 V. Specially, the efficiency of 1.93-eV single-junction and 3-J solar cells did not degrade in 300 s and 800 s under one-sun illumination, respectively. Importantly, the triple-junction device maintained 96% of its initial efficiency during MPPT in air for 1500 hours under simulated one-sun illumination. Overall, this work demonstrates an effective method to stabilize ultrawide-bandgap perovskite, paving the way for further performance improvements and the development of high-efficiency multijunction perovskite photovoltaics.

Methods

Materials

All materials were used as received without further purification. FAI, MAI, MABr and F-PEACl, CF₃-PACl were purchased from GreatCell Solar Materials. MACl and ethylenediamine diiodide (EDAI₂) were purchased from Xi'an Polymer Light Technology. CsI (99.9%) and CsBr (99.9%), 2PACz (>98.0%), MeO-2PACz (>98.0%), PbI₂ (99.99%), and PbBr₂ (99.99%) were purchased from Tokyo Chemical Industry (TCI). PEDAI (>98%) were purchased from Shanghai yuanye Bio-Technology Co., Ltd. 3MTPAI (99.5%) were purchased from Suzhou LiWei Technology Co., Ltd. DMF (99.8% anhydrous), DMSO (99.9% anhydrous) and anisole (99.8% anhydrous) were purchased from Sigma-Aldrich. MgF₂ (99.999%) was purchased from ZhongNuo Advanced Material Technology Co., Ltd. Tetrakis

(dimethylamino) tin(IV) (99.9999%) was purchased from Nanjing Ai Mou Yuan Scientific Equipment. ITO-glass has a square resistance of 20 ohm, purchased from SuZhou ShangYang Solar Technology.

Device fabrication:

For the 1.52-eV single-junction perovskite solar cells:

Patterned ITO glass substrates were sequentially cleaned using ultrasonication in acetone and isopropanol for 30 min, respectively. The cleaned ITO substrates were treated with ultraviolet ozone for 15 min, followed by sputtering a NiO_x hole transport layer on the substrates using a 3.5 inch NiO target (Plasmaterials, 99.9%) at 150 °C in an Ar atmosphere. The NiO_x layer was sputtered at a rate of 0.2 Å/s using a 90 W of RF power. Subsequently, the solutions of 2PACz and MeO-2PACz with the same concentration (1 mmol/L in IPA) were mixed with volume ratios (75:25), followed by washing with IPA for 2 times. FAI (219.17 mg), CsI (19.49 mg), MAI (23.84 mg), MACl (15 mg) and PbI₂ (691.2 mg) were dissolved in DMF: DMSO=800 ml : 200 ml. The 1.5M Cs_{0.05}FA_{0.85}MA_{0.1}PbI₃ perovskite precursor solution was spin-coated onto the ITO/NiO_x/SAM substrate at 1000 rpm for 10 s with an acceleration of 200 rpm and 6000 rpm for 40 s with an acceleration of 3000 rpm and 200 µl anisole was slowly dropped on the substrates at the last 15 s before the ending. The perovskite films were annealed at 100 °C for 45 min. For the device with passivation, the substrates were transferred into glovebox to spin-coating 1mg/ml F-PEACl and CF₃-PACl mixed solution with 4000 rpm., 30 s and annealed at 100 °C for 3 min. After that, 1mg/ml EDAI₂ was also spin-coated with the same processing program. All the passivation were dispersed in IPA as reported in previous work. Sequentially, 20 nm C₆₀ film was deposited by thermal evaporation at a rate of 0.2 Å s⁻¹. The substrates were then transferred to the atomic layer deposition (ALD) system (JIAXING KEMICRO TALD-200RR) to deposit 15 nm SnO₂ at low temperature (typically 100°C) using precursors of tetrakis (dimethylamino) tin(iv) and deionized water. Afterwards, 200 nm Ag as electrode was deposited by thermal evaporation at the rate of 1.0 Å s⁻¹.

For the 1.93-eV single-junction perovskite solar cells:

Patterned ITO glass substrates were sequentially cleaned using ultrasonication in acetone and isopropanol for 30 min, respectively. The cleaned ITO substrates were treated with ultraviolet ozone for 15 min. The NiO_x nanoparticles (15 mg ml⁻¹, H₂O) were spin-coated at 4000 rpm for 30 s, followed by annealing at 100 °C for 10 min in air. Subsequently, the solutions of 2PACz and MeO-2PACz with the same concentration (1 mmol/L in IPA) were mixed with volume ratios (75:25), followed by washing with IPA for 2 times. FAI (103 mg), CsI (65 mg), MABr (17 mg), PbBr₂ (304 mg) and PbI₂ (87.6 mg) were dissolved in DMF: DMSO=750 ml: 250 ml. The 1M Cs_{0.25}FA_{0.60}MA_{0.15}Pb(I_{0.43}Br_{0.57})₃ perovskite

precursor solution without/with 1mg PEDAI was spin-coated onto the ITO/NiO_x/SAM substrate at 1000 rpm for 10 s and 6000 rpm for 40 s with an acceleration of 3000 rpm and 200 μ l anisole was slowly dropped on the substrates at the last 20 s before the ending. The perovskite films were annealed at 100 °C for 15 min. For the device with passivation, the substrates were transferred into glovebox to spin-coating 0.5mg/ml 3MTPAI/IPA with 4000 rpm, 30 s and annealed at 100 °C for 10 min. Sequentially, 20 nm C₆₀ film was deposited by thermal evaporation at a rate of 0.2 Å s⁻¹. The substrates were then transferred to the atomic layer deposition (ALD) system (JIAXING KEMICRO TALD-200RR) to deposit 15 nm SnO₂ at low temperature (typically 100 °C) using precursors of tetrakis (dimethylamino) tin(IV) and deionized water. Afterwards, 200 nm of Ag as electrode was deposited by thermal evaporation at the rate of 1.0 Å s⁻¹.

Silicon bottom solar cell fabrication:

Silicon heterojunction (SHJ) bottom cells were fabricated on n-type double-side textured silicon wafers (CZ, 4 inch, 1-5 Ω resistivity, 260 μ m thickness). The intrinsic, n-type and p-type hydrogenated amorphous silicon layers were deposited on both sides of the wafer using plasma-enhanced chemical vapor deposition (PECVD). The back contact of the silicon cells was fabricated by stacking sputtering ITO (80 nm) and then thermal evaporated Ag (150 nm).

For the perovskite/perovskite/silicon 3-J solar cells:

Before the fabrication 1.52-eV perovskite top cells, the 4-inch silicon wafers were laser cut to 2.5 * 2.5 cm² substrates. Then the silicon substrates were annealed at 200 °C for 10 min to recover the sputtering damage. Next, 10 nm ITO was sputtered as recombination junction. The 1.52-eV perovskite top cells were fabricated using the identical process of the single-junction perovskite solar cells. Then, C₆₀ (15 nm) was deposited by a thermal evaporator. The following steps were taken: (1) For 2-J solar cells, SnO₂ (15 nm) was deposited by an atomic-layer deposition system, then the samples were transferred to a magnetron sputtering system to deposit 50 nm IZO, and a 400 nm Ag frame was deposited to complete the dual-junction device. (2) For 3-J cells, 60 nm of ALD-SnO₂ was deposited by an atomic-layer deposition system, 10 nm of ITO was sputtered by magnetron sputtering system. The 1.93-eV perovskite top cells were fabricated using the identical process of the single-junction perovskite solar cells. After the deposition of C₆₀ and ALD-SnO₂, 50 nm of IZO was sputtered. Ag (400 nm) were deposited by thermal evaporation at the rates of 1.0 Å s⁻¹. Finally, 100 nm of MgF₂ as antireflection layer was deposited by thermal evaporation.

Characterization of solar cells

For 3-J solar cells, J - V characteristics were measured using a Keithley 2400 source meter under the illumination of the solar simulator (EnliTech, Class AAA) at the light intensity of 100 mW cm^{-2} as verified with NREL calibrated reference solar cells (KG-5 and KG-0 reference cells were used for the measurements of wide-bandgap and narrow-bandgap solar cells, respectively). Unless otherwise stated, the J - V curves were all measured in air with a scan rate of 100 mV s^{-1} (voltage steps of 20 mV and a delay time of 100 ms). The active area was determined by the aperture shade masks (0.049 cm^2 and 1.17 cm^2) placed on the glass side of the solar cells. A dual-lamp solar simulator—rather than a single Xe lamp—is preferable for characterizing perovskite/perovskite/silicon 3-J tandem solar cells, as it offers a better spectral match with the AM1.5G solar irradiance (Supplementary Figures 41 and 42, see Supplementary Note 3 for details). EQE spectra were measured in ambient air using a QE system (EnliTech) with monochromatic light focused on device pixels and a chopper frequency of 20 Hz.

For 3-J solar cells, EQE measurements were performed in ambient air and the bias illumination from bright LEDs with emission peaks of 850 and 460 nm were used for the measurements of the front and back sub-cells, respectively. No bias voltage was applied during the EQE measurements of perovskite/perovskite/silicon solar cells. Light-emitting diode (LED) bias light (700 + 850 nm for 1.93-eV perovskite subcell, 460 + 850 nm for 1.52-eV perovskite subcell and 460 + 700 nm for silicon subcell). No bias voltage was applied during the EQE measurements of 3-J solar cells.

Other characterizations

SEM images were obtained using a TESCAN microscope with an accelerating voltage of 2 kV. XRD patterns were collected using a Bruker D8 Advance equipped with a NaI scintillation counter and using monochromatized Copper $K\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$). PL was measured using a HORIBA Nanolog fluorescence spectrometer (Horiba Instruments) with an excitation wavelength of 485 nm. The intensity of the laser was adjusted to 0.1-Sun equivalent. TRPL was measured using a Horiba Fluorolog-3 time-correlated single-photon counting system; the samples were excited using a pulsed laser with a wavelength of 485 nm from the perovskite side. The effective lifetime of perovskite films was calculated by fitting the biexponential components of PL decay curves. UPS with an excitation energy of 21.2 eV and an energy resolution of 50 meV was performed to obtain the secondary electron cut off and the valence band onset. XPS analysis was carried out using the Thermo Scientific Al K-alpha XPS system with energy steps of 0.1 eV.

Urbach energy were collected in ambient air using a FTSPS PECT-600 system with halogen lamp focused on the device pixel. The measurements were calibrated using Si and Ge reference cells.

Electrochemical impedance (EIS) was tested in the dark using an electrochemical workstation (Shanghai CH Instruments Co., Ltd), at room temperature with 40% humidity.

PLQY was obtained via exciting the corresponding samples with a 405 nm laser inside an integrated sphere (Newport, 70682NS). The signal was coupled into a fiber and collected by an Ocean Optics QE Pro spectrometer. The integrating sphere and fiber were calibrated using a lamp of the known spectrum (Ocean Insight, HL-3P-INT-CAL). The illumination intensity was modulated with neutral density (ND) filters to achieve a near 1-sun equivalent photon flux at a bandgap of 1.8 eV.

The TEM imaging was performed using a double aberration-corrected Titan G2 60–300 (S)TEM (Thermofisher Scientific Inc.) equipped with a X-FEG. HRTEM acquisition parameters, including spot size 7, magnification of 295 000, and appropriate monofocus and image corrector tuning were optimized to achieve an electron dose rate of $\approx 40 \text{ e } \text{\AA}^{-2}\text{s}^{-1}$. A 16-megapixel fast CMOS camera (Oneview, Gatan) was utilized for HRTEM acquisition, with a single frame exposure time of 1 s. The perovskite film on the glass is scraped off and placed in a centrifuge tube, and ultrasonically dispersed in IPA solvent. Hold the copper mesh with self-locking tweezers and suspend it in the air. Afterward, we dropped 5 μL of the as-prepared solution on copper mesh with microgrid for the test of TEM. This process is repeated multiple times to ensure that sufficient amounts of the film are deposited on the copper mesh and annealed. Such a dose is sufficient for imaging the sample without causing damage from electron radiation.

Data availability

Source data are provided with this paper. The main data supporting the findings of this study are available within the published article and its Supplementary Information and source data files. Additional data are available from the corresponding author on request.

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Author contributions

Conceptualization, W.K., X.H., H.L., and H.T.; investigation, W.K., X.H., H.L., B.Y., J.S.; resources, Y.R., R.Li., D.Y., Y.L., J.W., J.H., X.Z., H.F., L.Z., Z.C., R.L., Y.D., and H.T.; validation, W.K.; visualization, W.K.; writing – original draft, W.K.; writing – review & editing, X.H., H.L., B.Y., J.S., and H.T.; supervision, W.K., and H.T.; project administration, W.K., and H.T.; funding acquisition, W.K., and H.T.

Competing interests

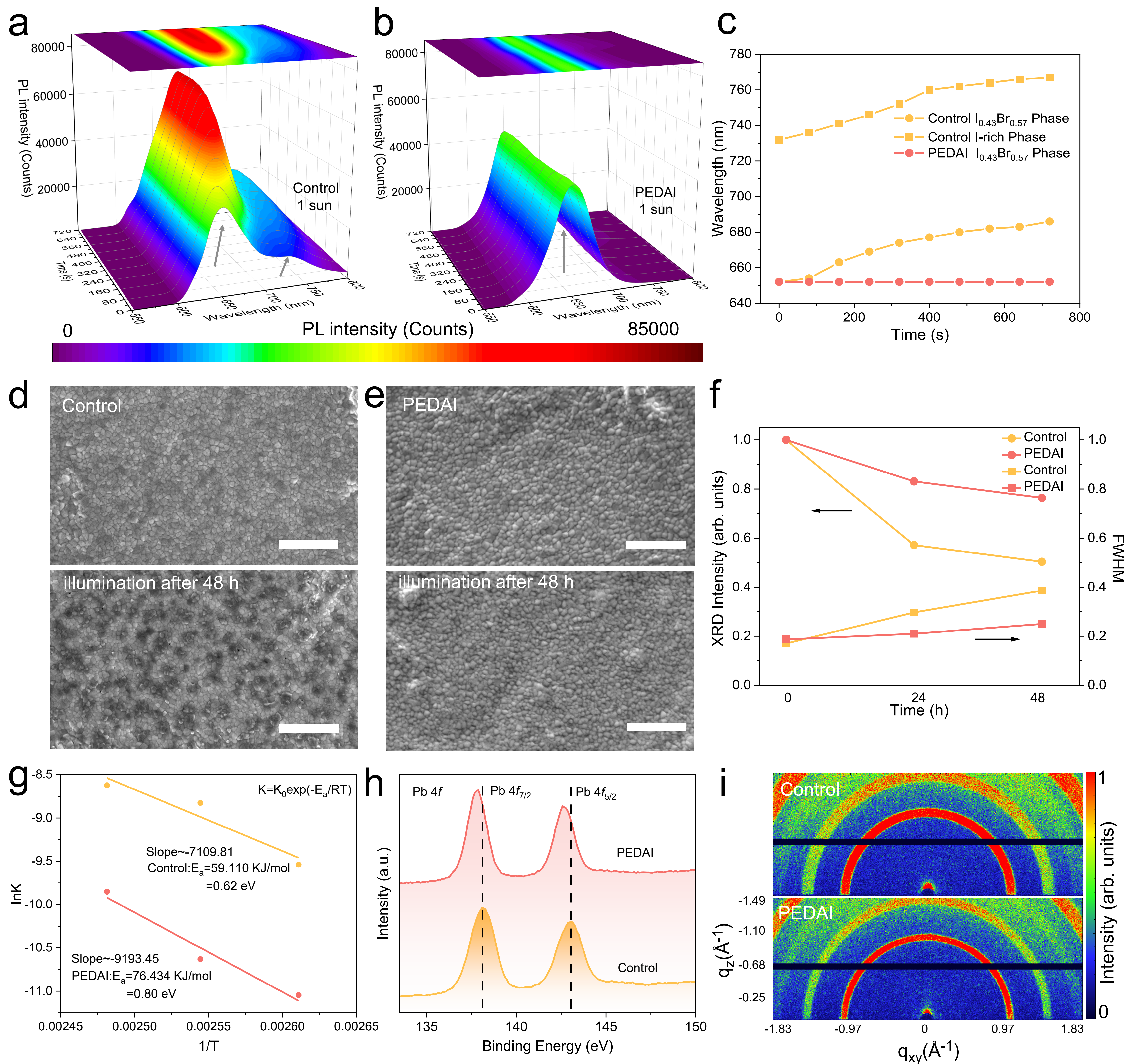
Hairen Tan is the founder, Chief Scientific Officer, and Chairman of Renshine Solar Co., Ltd., a company that is commercializing perovskite PVs. The remaining authors declare no competing interests.

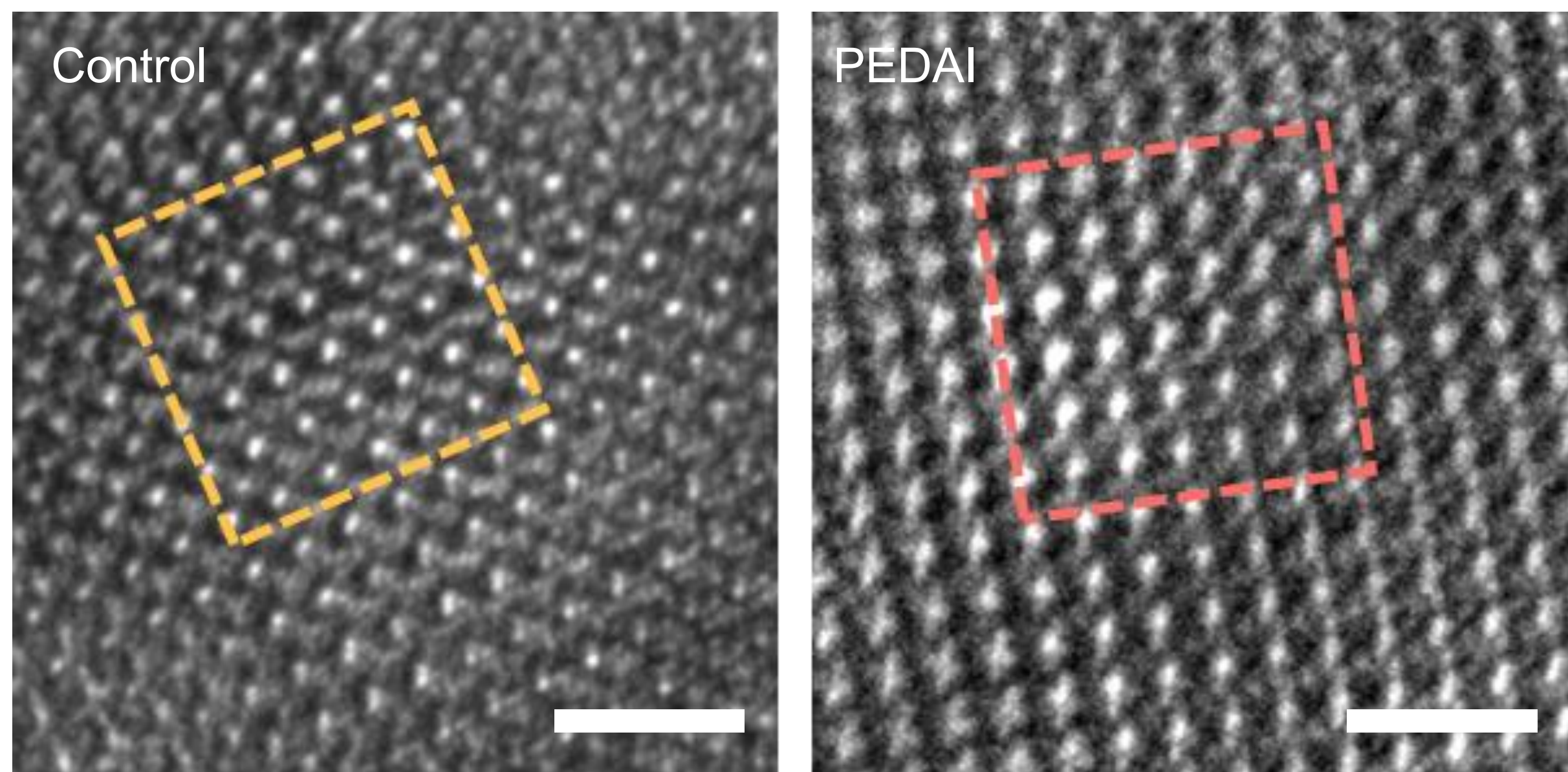
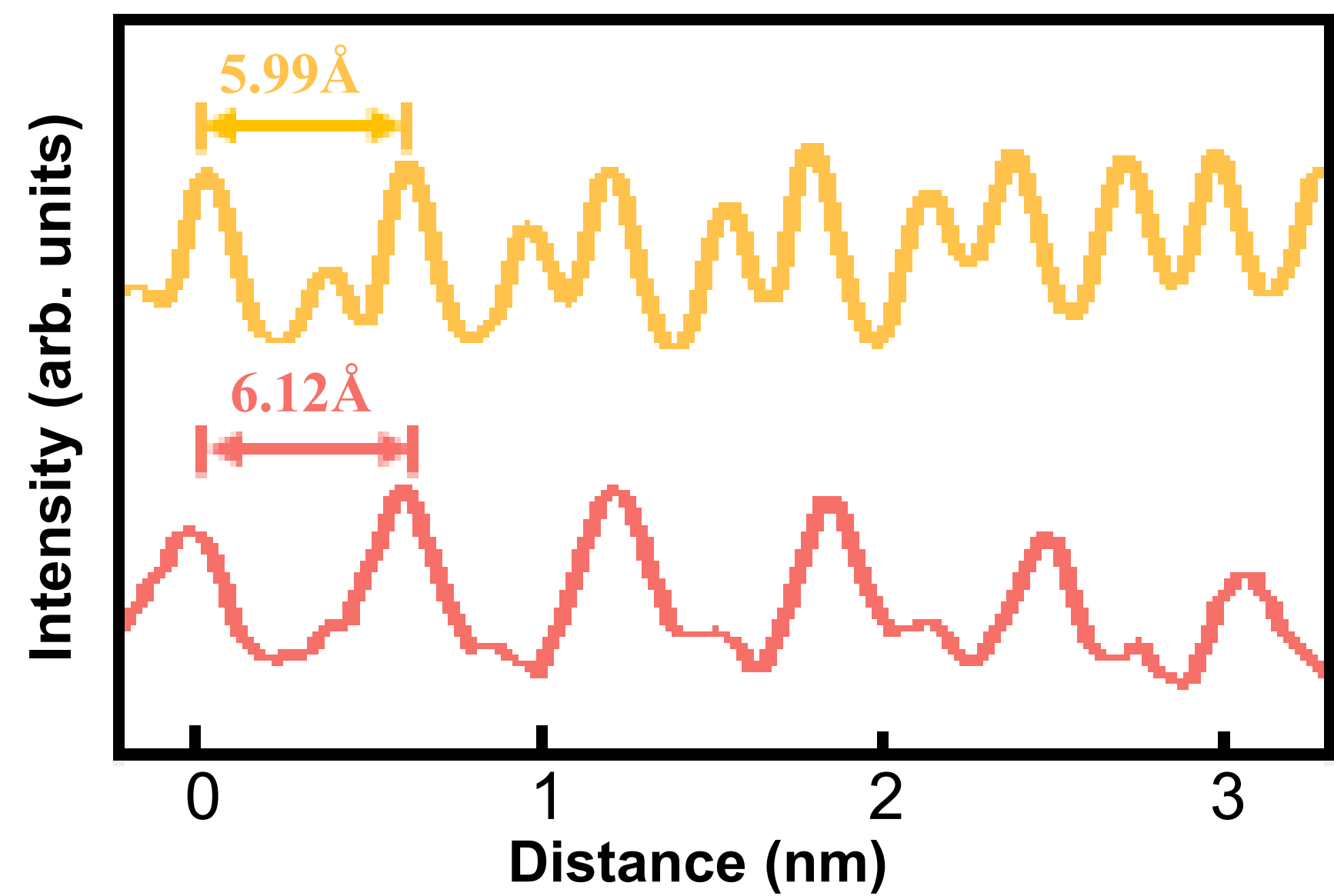
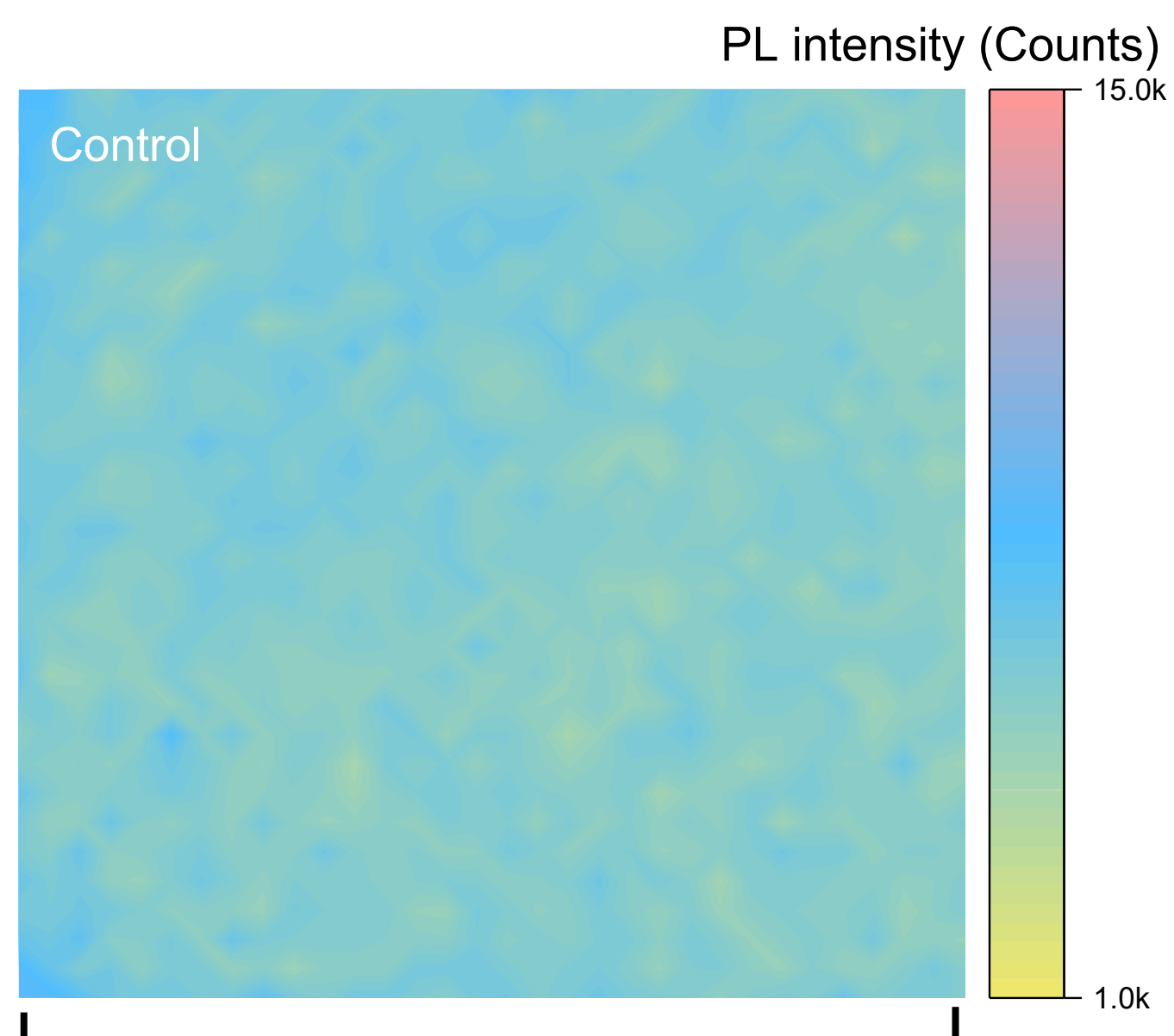
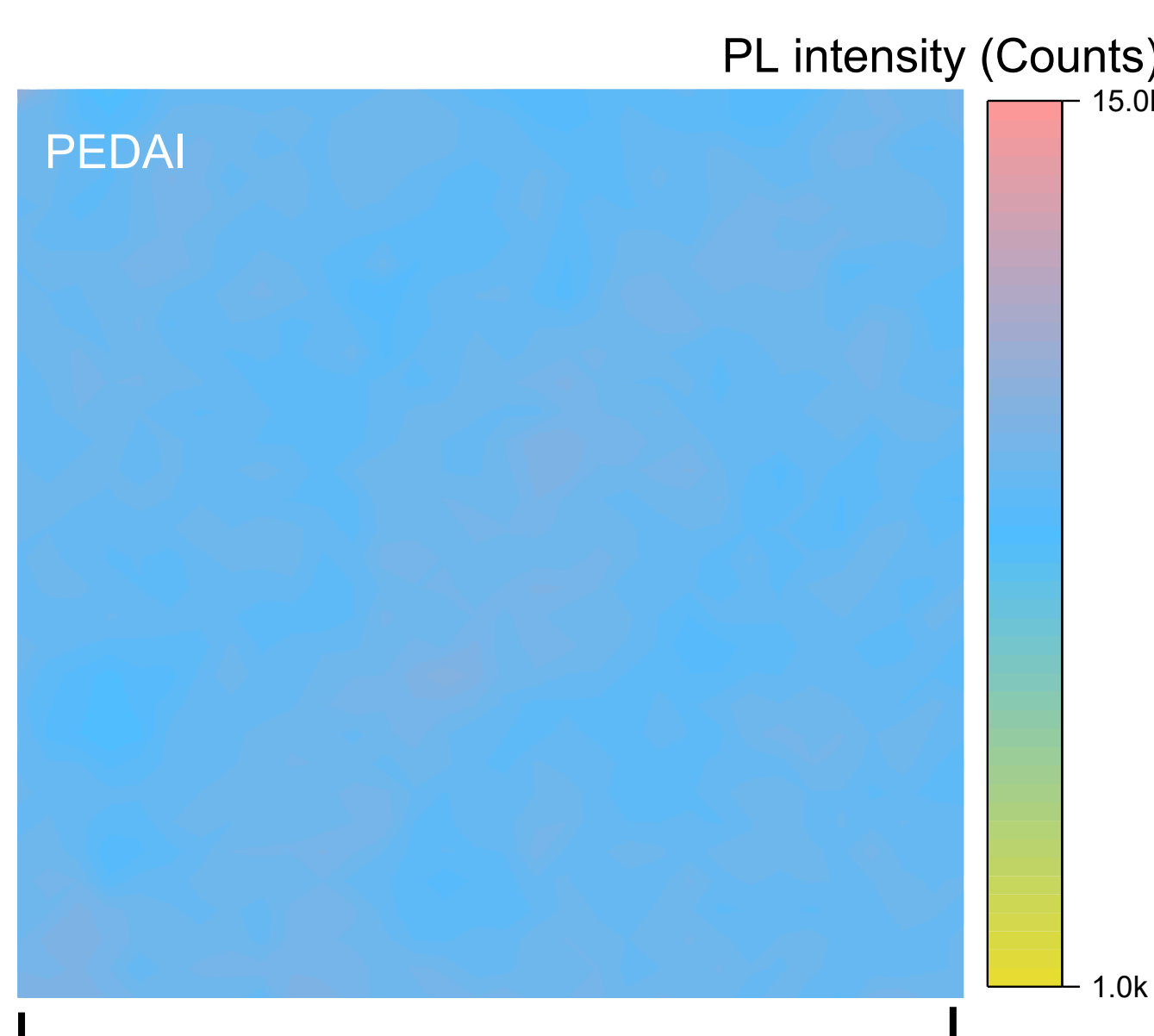
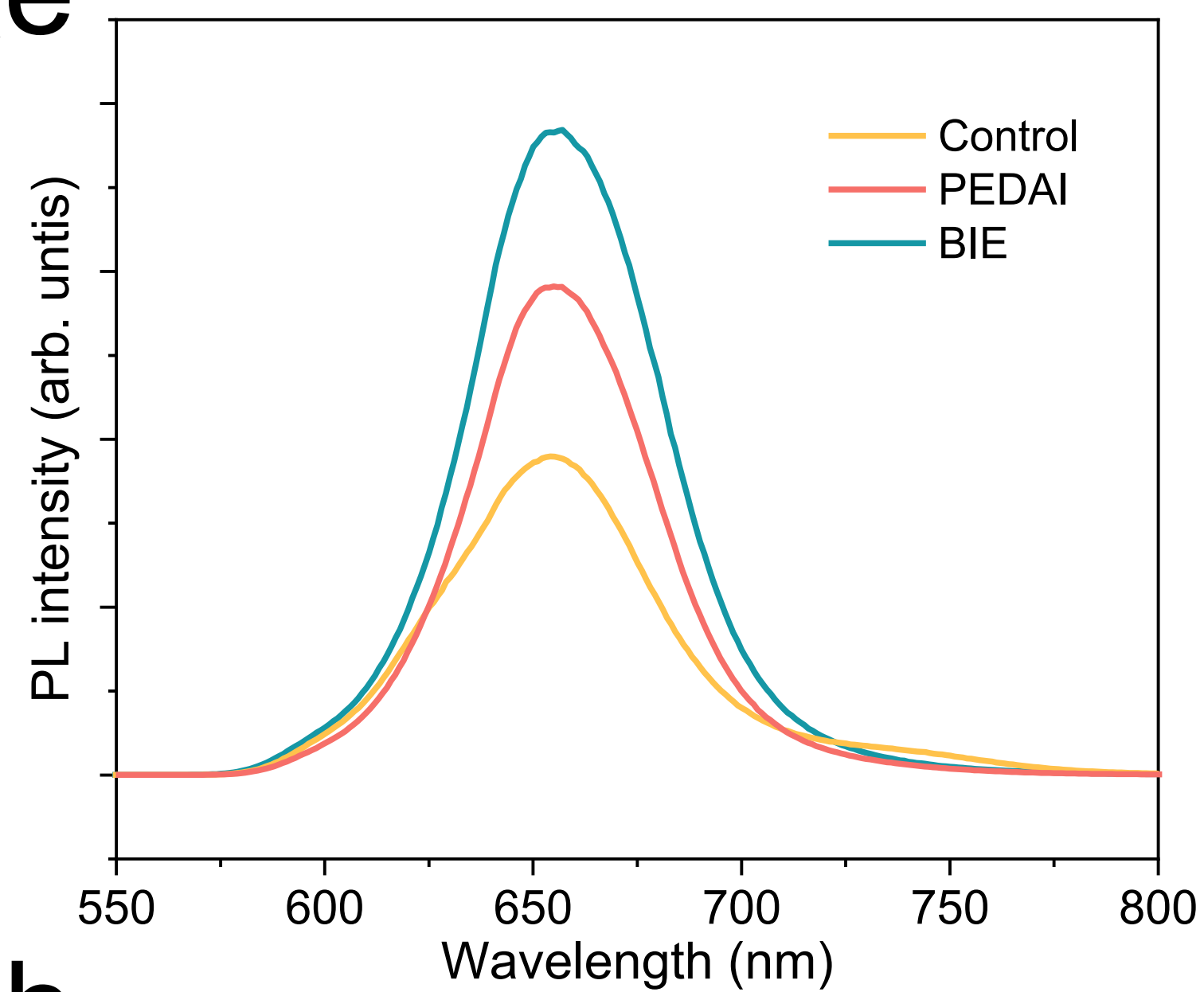
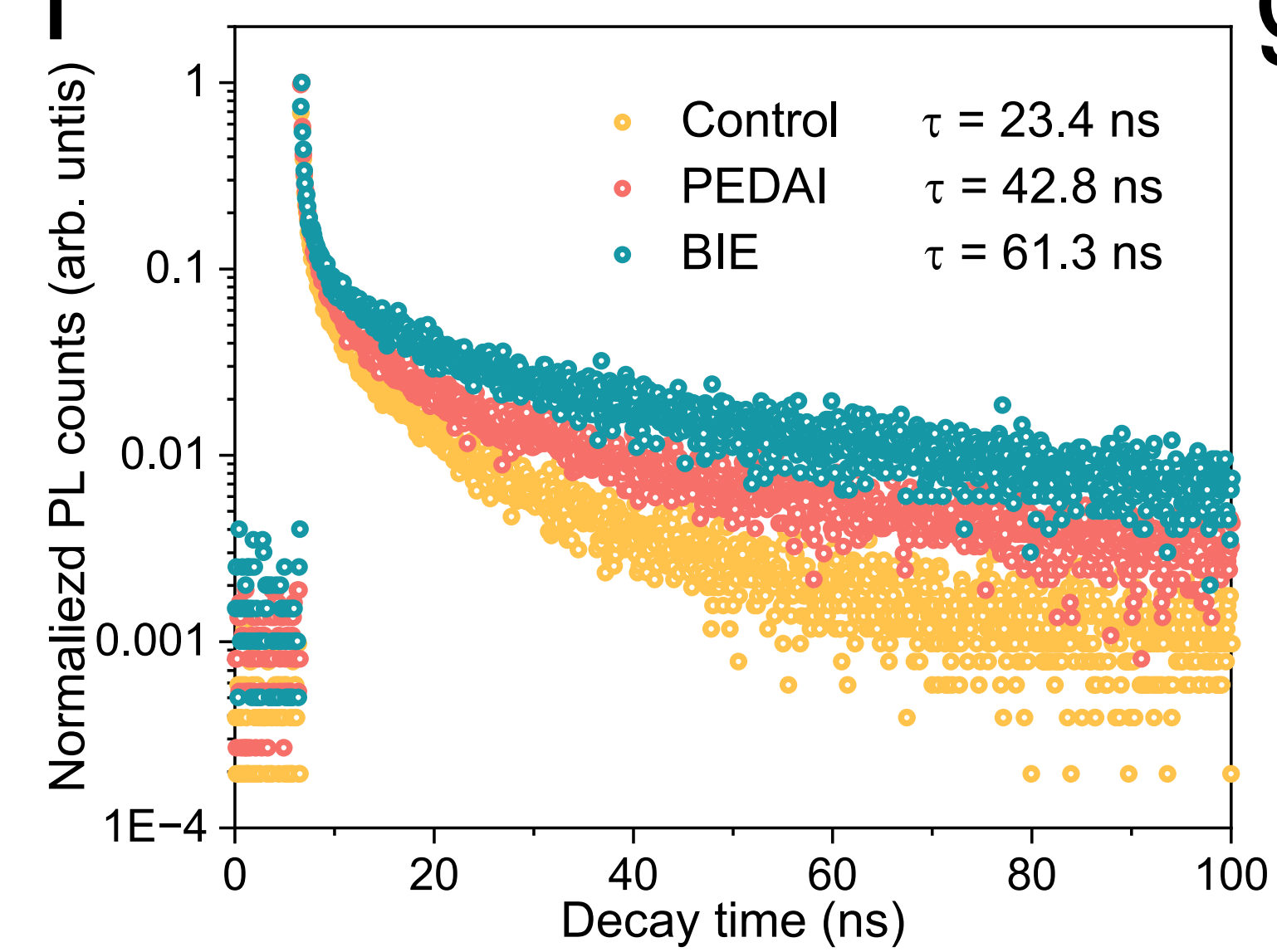
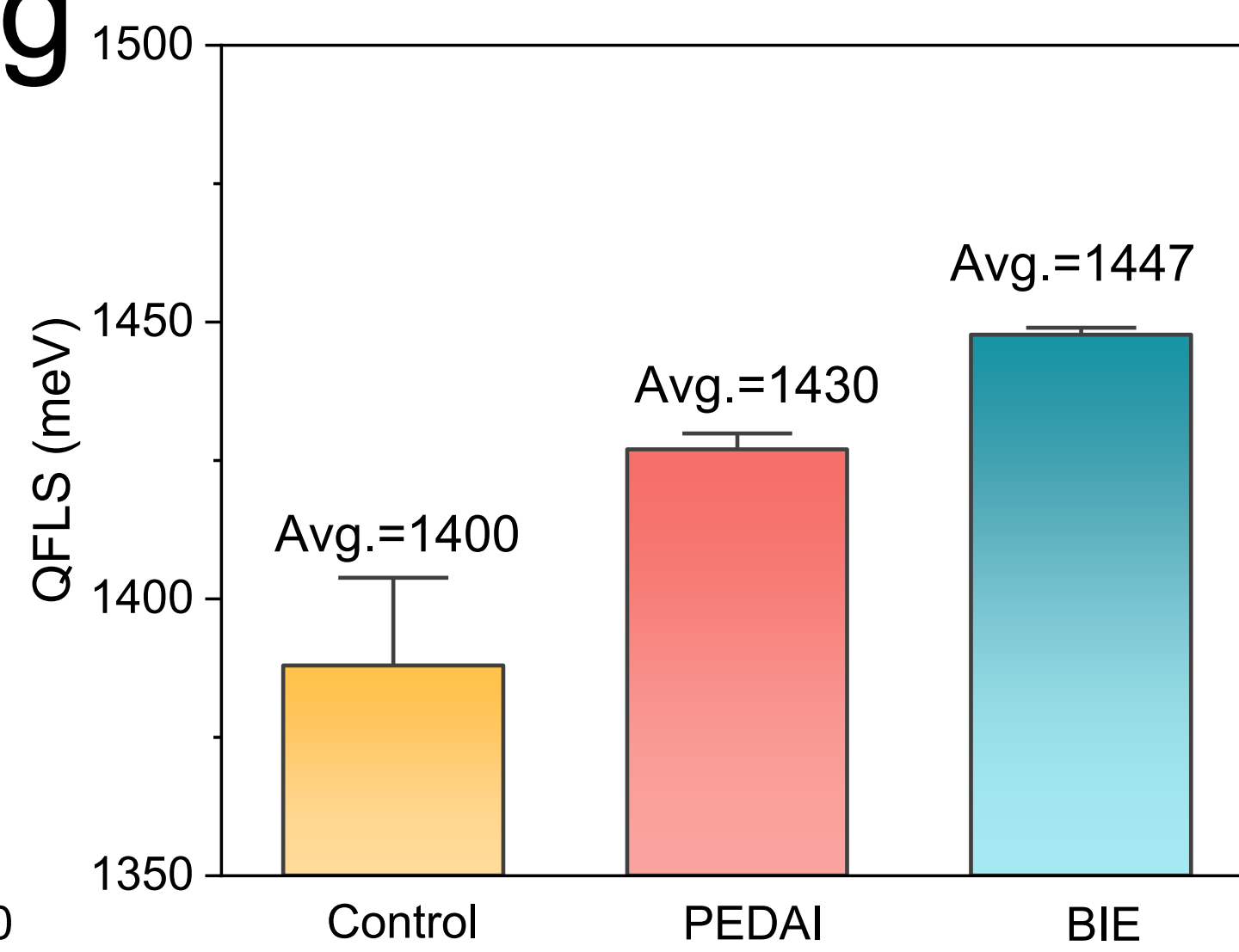
Fig. 1 | Optical, morphology and structural evolution of the control and PEDAI ultrawide-bandgap perovskite films under light condition. Evolution of the PL spectrum over time for **a** control, and **b** PEDAI films under continuous light 720 s. The colour scale shows intensity from low (light purple) to high (dark red). **c** Time-dependent PL wavelength of control and PEDAI films. Top-view SEM images of **d** control and **e** PEDAI films before and after 48 h under 1-sun illumination in glovebox. Scale bar, 2 μm . **f** Time-dependent XRD patterns fresh and aged under 1-sun illumination in glovebox. **g** Arrhenius plot and extracted activation energy for the control and PEDAI films. **h** The XPS pattern of Pb 4f core levels of the control and PEDAI films, respectively. **i** GIWAXS patterns of the control and PEDAI film. The colour scale shows intensity from low (black) to high (red).

Fig. 2 | Photostability and charge-carrier dynamics of the control and PEDAI films. **a** TEM images of the control and PEDAI films, and **b** the profile and magnitude of the interplanar spacing corresponding to the square area. Scale bar, 2 nm. **c** PL mapping of the control, and **d** PEDAI films with the active area for 1.5 cm * 1.5 cm. The colour scale shows intensity from low (yellow) to high (red). **e** PL spectra, **f** TRPL spectra and **g** QFSL of the control, PEDAI and BIE films. The standard deviation has been employed to assess data dispersion in statistical analyses. **h** Schematic energy level alignment extracted from UPS data between the control, PEDAI, BIE films and C₆₀. All films were deposited on glass.

Fig. 3 | Characterization of photovoltaic performance for 1.93-eV single-junction devices. **a** Photovoltaic parameters V_{oc} , FF, J_{sc} and PCE of the control, PEDAI and BIE devices in the same runs (0.049 cm^2). The box lines indicate the standard deviation, and the centre represents the mean value. The standard deviation has been employed to assess data dispersion in statistical analyses. **b** J - V curves, **c** EQE spectra and **d** Stabilized power output of the champion control, PEDAI and BIE devices. The inset in **d** is a photo of the 1.93-eV single-junction device. **e** V_{oc} evolution as a function of light intensity for the control, PEDAI and BIE devices. **f** The E_U and **g** the EIS (generated using Origin 2021 software) of the control, PEDAI and BIE devices. The standard deviation has been employed to assess data dispersion in statistical analyses.

Fig. 4 | Photovoltaic performance of monolithic perovskite/perovskite/silicon triple-junction solar cells. **a** Schematic diagram of perovskite/perovskite/silicon solar cells. This schematic diagram was created using Microsoft Office PowerPoint 2021. **b** Cross-section SEM image of the fabricated 3-J solar cells. Scale bar, 1 μm . **c** The digital photos of 3-J device, area: 2.5 cm * 2.5 cm. **d** $J-V$ and **e** EQE curves of the champion triple-junction device (1.17 cm^2 aperture area). **f** MPPT measurement of the unencapsulated champion triple-junction device for 800 s under continuous 1-sun illumination in air. **g** MPPT of encapsulated BIE 3-J devices under continuous 1-sun illumination with AM 1.5G full spectrum (ISOS-L-1). **h** Summary of the reported MPPT of perovskite/perovskite/silicon and perovskite/perovskite/perovskite triple-junction solar cells based on ultrawide-bandgap.



a**b****c****d****e****f****g****h**